Completion of the singleton and unique-crossing branches

Human Compression project: L044 continuation

Let $G \leq \mathrm{Alt}(\Omega)$ be transitive with a connected string C-marking $\rho_0, \dots, \rho_{r-1}$, and suppose every maximal ambient omission is intransitive. Write $n = |\Omega| \geq 12$. Assume the alternating rank bound at every strictly smaller degree at least twelve. All complete acquired primitive, anchored-imprimitive, small-degree and attachment inputs are retained with their actual scope; this induction premise does not presume the theorem at degree n. Eligibility means crossing two actual ambient omission orbits.

Theorem 1 (A singleton at an end)**.** *If G_0 has the two orbits $\{a\}$ and $\Omega\setminus\{a\}$, then $r \leq \lfloor (n-1)/2 \rfloor$. The same conclusion holds with the string reversed.*

Proof. Suppose the bound fails. The complete critical odd-tail reduction gives $n = 2r \geq 14$ and the actual natural $G_0 = \mathrm{Alt}(\Omega \setminus \{a\})$. The complete critical-head theorem excludes either actual supported six-point A_5 end from source Table 2(2)–(3), applying full string reversal at the other end. Its small boundary and the unextendable orientation are retained in that theorem's proof.

The complete singleton global-pairs theorem supplies a further colour $j \geq 2$ with exactly one ambient-eligible edge. Choose the least such index. Every colour $2, \dots, j-1$ is paired: it has at least one eligible edge, since otherwise adjoining its mark could not make its intransitive omission transitive, and minimality excludes exactly one.

By the actual singleton-prefix lemma the two G_j-orbits are

$$C = a^{G_{<j}}, \qquad A = \Omega \setminus C,$$

the suffix $G_{>j}$ fixes C pointwise, and every shared label at this cut is below j. Strict growth gives $|C| \geq j+1$.

First suppose $A = \{d\}$ is a singleton. If $j < r-1$, the complete interior singleton-orbit theorem gives the required bound, contrary to our assumption. If $j = r-1$, reverse the whole string and apply the critical odd-tail reduction to this actual endpoint omission. It gives $G_{r-1} = \mathrm{Alt}(\Omega\setminus\{d\})$. The minimal choice of j supplies every paired colour $2, \dots, r-2$. These are precisely the hypotheses of the complete two-singleton theorem, which excludes the configuration. Thus A is nontrivial from now on.

If no label is shared across the cut, each nonidentity mark other than ρ_j acts on exactly one of C, A. Within either of the two string intervals the side cannot change: otherwise two adjacent marks would have disjoint supports and commute, disconnecting the full string diagram. The prefix acts on C, since ρ_0 moves a. Nontriviality of A forces $j < r-1$ and makes the suffix act on A. Thus these are actual supported transitive prefix and suffix groups, their transitivity following from the two actual omission orbits. The complete supported-interval alternative gives the rank bound or an actual exceptional end, both already excluded.

Consequently a first shared label $g < j$ exists. The complete singleton-prefix support theorem gives

$$g \geq 3, \qquad Y = \mathrm{supp}(G_{<g}) \subseteq C, \qquad |Y| \leq 2g+1. \tag{1}$$

Here the support inclusion is actual: every earlier unshared mark acts on the same side, namely C, because the prefix starts at ρ_0. The small cases $g = 1, 2$ have already been excluded in that theorem, including its full critical-dihedral argument at the original reversed cut.

If $G_{<g}$ is transitive on Y, use the full transitive-tail alternative after reversing the entire string. Its crossing is $i = r-1-j$ and its last shared label is $h = r-1-g$, so $i < h < r-1$. Its

supported tail past h is exactly the actual $G_{<g}$ on $Y \subseteq C$. If $i > 0$, the preceding subgroup in this orientation is the original $G_{>j}$, which fixes C pointwise. Thus the chosen-orientation condition is proved, not inferred from a projected marking. If $i = 0$, the endpoint form applies. In both cases the theorem gives the rank bound or an actual exceptional end, again impossible.

We are left with $G_{<g}$ intransitive on its support. At $g = 3$, (1) contradicts the complete third-prefix hand proof, which requires at least eight moved points. For every $g \geq 4$, put $H = G_{\{2,\dots,g-1\}}$. All these colours are paired by $g < j$ and the least choice of j. The complete small-support budget therefore applies with its exact actual hypotheses. If f counts the H-fixed points in Y and c its nontrivial orbits, its exhaustive table is

$$(f, c, |Y|) \in \{(2, 2, 2g), (2, 2, 2g + 1), (3, 2, 2g + 1), (2, 3, 2g + 1)\}.$$

The complete two-orbit theorem excludes the first two entries; the third-fixed-point theorem excludes the third; the selected ambient confinement and three-orbit theorem exclude the fourth. All hypotheses concern the actual prefix and actual ambient eligible edges. No pure alternating prefix kernel or projected intersection property is used.

This exhausts every later cut and every possible first shared prefix. The assumed rank failure is impossible. □

Corollary 2 (The unique-crossing branch)**.** *Under the opening hypotheses, if at least one colour has exactly one ambient-eligible edge, then* $r \leq \lfloor (n-1)/2 \rfloor$.

Proof. Suppose the bound fails and choose such a crossing of index i. Its omission has exactly two orbits. If either is a singleton, the interior singleton-orbit theorem or Theorem 1, with full reversal when necessary, excludes it.

Suppose both orbits are nontrivial. If no label is shared, the same connected-interval argument used above puts all marks on opposite sides of the crossing. The crossing is interior, since at an endpoint its only remaining interval could act on just one orbit, leaving the other a singleton. The supported-interval alternative therefore applies and gives an exceptional actual end, excluded by the complete critical-head theorem.

If some label is shared, reverse the full string if necessary so that at least one unique crossing has a shared label to its right. Choose the maximal index among all such crossings and apply the complete maximal-crossing alternative. It gives the rank bound, an actual exceptional end, or the terminal singleton case. Its retained proof obtains this last case at an actual later *unique* crossing $j = r - 1$; hence that endpoint omission has exactly the singleton and its complement as its two orbits. This is the full two-orbit hypothesis needed by Theorem 1, not merely an arbitrary fixed point of an omission. Each alternative is now excluded. □

Attribution and principal-source boundary. The cut strategy and rank target are those of Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for* A_n, arXiv:1605.09173v1, Section 5. This assembly retains all complete acquired proofs, including explicit smaller-degree induction and their required finite inputs. The new uniform prefix orbit exclusions supply the missing singleton continuation. The separately reviewed fourth-prefix GAP certificate is unnecessary as an additional premise here, since the uniform budget covers every $g \geq 4$.

The conclusion covers the all-intransitive-omission branch with at least one unique crossing. The all-paired equality case remains separate, as do general transitive imprimitive maximal omissions, the exact base maxima, lower-existence bindings and the complete principal-theorem assembly. No whole L044 recovery, conjecture, measurement or new formal declaration is claimed.